(1) a)

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-g(x_i)|^2} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-y_i+y_i-g(x_i)|^2}$$

$$\leq \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(|f^*(x_i)-y_i| + |y_i-g(x_i)|\right)^2} =$$

$$= \sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-y_i|^2 + \frac{1}{n}\sum_{i=1}^{n}|y_i-g(x_i)|^2 + \frac{2}{n}\sum_{i=1}^{n}|f^*(x_i)-y_i||y_i-g(x_i)|}$$

$$\leq \sqrt{\left(\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-y_i|^2} + \sqrt{\frac{1}{n}\sum_{i=1}^{n}|y_i-g(x_i)|^2}\right)^2} = \underbrace{\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-y_i|^2}}_{\leq \varepsilon} + \sqrt{\frac{1}{n}\sum_{i=1}^{n}\underbrace{|y_i-g(x_i)|}_{e_i}{}^2}$$

by Cauchy Schwartz:

$$\sum_{i=1}^{n}|u_i||v_i| \leq \sqrt{\sum_{i=1}^{n}|u_i|^2}\sqrt{\sum_{i=1}^{n}|v_i|^2}$$

$$\leq \varepsilon + \sqrt{\frac{1}{n}\sum_{i=1}^{n}\underbrace{|e_i|^2}_{\leq E}} \leq \varepsilon + \sqrt{\frac{1}{n}\sum_{i=1}^{n}E^2} = \varepsilon + E, \text{ and letting}$$

$h(E) = E$, we get:

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-g(x_i)|^2} \leq \varepsilon + h(E) = \varepsilon + E, \text{ which}$$

demonstrates such $h(E)$ independent of ε exists $\square$.

b) Let $x, y \in \mathbb{R}^d$, then:

$$|\phi((W_j^*)^T x + b_j^*) - \underbrace{\phi}_{\max\{0, \text{input}\}}((W_j^*)^T y + b_j^*)| =$$

$$= |\max\{0, (W_j^*)^T x + b_j^*\} - \max\{0, (W_j^*)^T x + b_j^*\}|.$$

Next consider: $\max\{0,a\} - \max\{0,b\}$, 4 cases:

$a<0, b<0$: $|\max\{0,a\} - \max\{0,b\}| = 0 \leq |a-b|$ ← always ≥ 0

$a>0, b<0$: $|\max\{0,a\} - \max\{0,b\}| = |a| \leq |a-b|$

$a<0, b>0$: $|\max\{0,a\} - \max\{0,b\}| = |-b| \leq |b-a| = |a-b|$ (since $-b \geq 0$)

$a>0, b>0$: $|\max\{0,a\} - \max\{0,b\}| = |a-b| \leq |a-b|$ ($-a \geq 0$). So it holds:

$|\max\{0,a\} - \max\{0,b\}| \leq |a-b|$.

Therefore:

$$|\phi((W_j^*)^T x + b_j^*) - \phi((W_j^*)^T y + b_j)| \leq |(W_j^*)^T x + b_j - ((W_j^*)^T y + b_j)|$$

$$= |(W_j^*)^T (x-y)| = \left|\sum_{k=1}^{d} W_{j,k}^* (x_k - y_k)\right| \leq \sum_{k=1}^{d} |W_{j,k}^*||x_k - y_k|$$

$$= \sum_{k=1}^{d} |W_{j,k}^*| \sqrt{(x_k - y_k)^2} \leq \sum_{k=1}^{d} |W_{j,k}^*| \underbrace{\sqrt{\sum_{i=1}^{d} (x_k - y_k)^2}}_{\text{L-2 norm}} = \underbrace{\left(\sum_{k=1}^{d} |W_{j,k}^*|\right)}_{:=L} \|x-y\|_2$$

$= L\|x-y\|_2$. So $\phi((W_j^*)^T x + b_j)$ is Lipschitz by definition, with Lipschitz constant $L = \left(\sum_{k=1}^{d} |W_{j,k}^*|\right)$.

(1) c) The original relu is: $\phi(z) = \max(0,z)$. Then, define:

$$\phi_\sigma(z) = \int_{-\infty}^{\infty} \phi(z-t)\, p_\sigma(t)\, dt \underset{\text{by definition}}{=} \mathbb{E}\left[\phi(z-Z)\right]$$

($p_\sigma(t)$: gaussian pdf; Z: gaussian RV.)

i.e. $p_\sigma(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{t^2}{2\sigma^2}\right\}$

Next: $|f^*(x_i) - f^*_\sigma(x_i)| = |(V^*)^T \phi(W^* x_i + b^*) - (V^*)^T \phi_\sigma(W^* x_i + b^*)|$

$= \left|(V^*)^T \left(\phi(W^* x_i + b^*) - \mathbb{E}\left[\phi(W^* x_i + b^* - Z)\right]\right)\right| =$

$= \left|(V^*)^T \mathbb{E}\left[\phi(W^* x_i + b^*) - \phi(W^* x_i + b^* - Z)\right]\right| =$

$= \left|\sum_{j=1}^{m} (V_j^*)\, \mathbb{E}\left[\phi(W^* x_i + b^*) - \phi(W^* x_i + b^* - Z)\right]_j\right| \underset{\text{triangle ineq and } |\mathbb{E}[X]| \le \mathbb{E}|X|}{\le} \sum_{j=1}^{m} |V_j^*|\, \mathbb{E}\left|\phi(W^* x_i + b^*) - \phi(W^* x_i + b - Z)\right|_j$

$\underset{\text{from before}}{\le} \sum_{j=1}^{m} |V_j^*|\, \underbrace{\mathbb{E}|Z|_j}_{\text{gaussian RV}} = \sum_{j=1}^{m} |V_j^*|\, \underbrace{\sigma\sqrt{\frac{2}{\pi}}}_{\mathbb{E}|Z| \text{ when } Z\sim N(0,\sigma^2)} = \sigma\sqrt{\frac{2}{\pi}}\, \|V^*\|_1 \underset{\|V^*\|_1 \le \sqrt{m}\,\|V^*\|_2 \text{ by Cauchy-Schwartz}}{\le} \sigma\sqrt{m}\sqrt{\frac{2}{\pi}}\, \|V^*\|_2$

So: $|f^*(x_i) - f^*_\sigma(x_i)| \le \sigma\sqrt{\frac{2m}{\pi}}\, \|V^*\|_2$.

Then, since this holds for all x_i:

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n} |f^*(x_i) - f^*_\sigma(x_i)|^2} \le \sqrt{\frac{1}{n}\cdot \sum_{i=1}^{n} \frac{\sigma^2 \cdot 2m}{\pi}\, \|V^*\|_2^2} = \sigma\sqrt{\frac{2m}{\pi}}\, \|V^*\|_2,$$

establishing the desired bound (i.e. a valid upper bound).

d) First, by a) and c):

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f_\sigma^*(x_i)-g(x_i)|^2} \le \underbrace{\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f_\sigma^*(x_i)-f^*(x_i)|^2}}_{\le \sigma\sqrt{2m/\pi}\,\|v^*\|_2} + \underbrace{\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f^*(x_i)-g(x_i)|^2}}_{\le \varepsilon + h(E)}$$

$$\le \sigma\sqrt{\frac{2m}{\pi}}\,\|v^*\|_2 + \varepsilon + \underbrace{h(E)}_{=E}.$$

Now set: $\sigma\sqrt{\frac{2m}{\pi}}\,\|v^*\|_2 + \varepsilon + E \le 2\varepsilon \Leftrightarrow \sigma\sqrt{\frac{2m}{\pi}}\,\|v^*\|_2 + E \le \varepsilon.$

Let: $E \le \frac{\varepsilon}{2}$, $\sigma\sqrt{\frac{2m}{\pi}}\,\|v^*\|_2 \le \frac{\varepsilon}{2} \Rightarrow \sigma \le \frac{\varepsilon}{2\|v^*\|_2}\sqrt{\frac{\pi}{2m}}$.

So the following upperbounds: $E \le \frac{\varepsilon}{2}$, $\sigma \le \frac{\varepsilon}{2\|v^*\|_2}\sqrt{\frac{\pi}{2m}}$ are sufficient to ensure:

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}|f_\sigma^*(x_i)-g(x_i)|^2} \le 2\varepsilon.$$

(2) a)

$$\mathbb{E}_{u_1,\dots,u_N}[\tilde{h}(x)] = \mathbb{E}_{u_1,\dots,u_N}\left[\frac{1}{N}\sum_{j=1}^{N}\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle)\right] =$$

$$= \frac{1}{N}\sum_{j=1}^{N}\mathbb{E}_{\substack{\vec{u}\\ \vdots \\ u_1,\dots,u_N}}\left[\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle)\right] \underset{\text{iid}}{=} \mathbb{E}_{\vec{u}}\left[\tilde{w}(u)\,\sigma(\langle u,x\rangle)\right] \text{ for } u\sim p_u$$

u_j's are iid, so each expectation is the same

By definition:

$$\mathbb{E}_{u_1,\dots,u_N}[\tilde{h}(x)] = \mathbb{E}_{\vec{u}}\left[\tilde{w}(u)\,\sigma(\langle u,x\rangle)\right] = \int_u p_u(u)\,\tilde{w}(u)\,\sigma(\langle u,x\rangle)\,du =$$

(we want this to hold)

$$h(x) = \int_u d(u)\,\sigma(\langle u,x\rangle)\,du \Leftrightarrow p_u(u)\,\tilde{w}(u) = d(u) \Leftrightarrow$$

$$\tilde{w}(u) = \frac{d(u)}{p_u(u)} \quad \text{where } p_u \text{ is the probability density.}$$

Now, let $\tilde{w}(u) = \frac{d(u)}{p_u(u)}$. Then:

$$\mathbb{E}_{u_1,\dots,u_N}\left[|\tilde{h}_u(x) - h(x)|\right] = \int_u p_u(u)\sqrt{(\tilde{h}_u(x)-h(x))^2}\,du = \int_u \underbrace{\sqrt{p_u(u)}}_{d(u)}\underbrace{\sqrt{p_u(u)(\tilde{h}_u(x)-h(x))^2}}_{f(u)}\,du$$

$$\underset{\text{Cauchy Schwarz}}{\leq} \sqrt{\int_u p_u(u)(\tilde{h}_u(x)-h(x))^2\,du}\;\underbrace{\sqrt{\int_u p_u(u)\,du}}_{=1} = \sqrt{\mathbb{E}_{u_1,\dots u_N}[(\tilde{h}_u(x)-h(x))^2]} =$$

$\int f(u)g(u)\,du \leq \sqrt{\int f(u)^2\,du}\sqrt{\int g(u)^2\,du}$

$$= \sqrt{\mathbb{E}_{u_1,\dots,u_N}\left[\left(\frac{1}{N}\sum_{j=1}^{N}\left(\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle)-h(x)\right)\right)^2\right]} = \sqrt{\frac{1}{N^2}\mathbb{E}_{\vec{u}}\left[\left(\sum_{j=1}^{N}\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle)-h(x)\right)^2\right]}$$

$$= \frac{1}{N}\sqrt{\sum_{j=1}^{N}\mathbb{E}_{\vec{u}}\left[\left(\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle)-h(x)\right)^2\right]}$$

the sum goes out because $\tilde{w}(u_j)\,\sigma(\langle u_j,x\rangle) \perp \tilde{w}(u_k)\,\sigma(\langle u_k,x\rangle)$ for $j\neq k$, so by unbiasedness $\mathbb{E}=0$.

to clarify $\mathbb{E}[(\sum W_j)^2] = \mathbb{E}[\sum_{j=1}^{N} W_j^2] + \mathbb{E}[\sum_{j\neq k} W_j V_k]$,

but for $j\neq k$: $\mathbb{E}[\sum_{j\neq k} W_j W_k] = \sum_{j\neq k}\underbrace{\mathbb{E}[W_j]}_{0}\underbrace{\mathbb{E}[W_k]}_{0} = 0$.

Next, note that: $\mathbb{E}_{\vec{w}}\left[\left(\tilde{w}(u_j)\,\sigma(\langle u_j, x\rangle) - h(x)\right)^2\right] =$

$= \mathbb{E}_{\vec{w}}\left[\left(\tilde{w}(u_j)\,\sigma(\langle u_j, x\rangle)\right)^2\right] - 2\underbrace{\mathbb{E}_{\vec{w}}\left[w(u_j)\,\sigma(\langle u_j, x\rangle)\right]}_{= \mathbb{E}_{\vec{w}}[\hat{h}(x)] = h(x)} h(x) + h(x)^2 =$

$= \mathbb{E}_{\vec{w}}\left[\left(\tilde{w}(u_j)\,\sigma(\langle u_j, x\rangle)\right)^2\right] - h(x)^2 \leq \mathbb{E}_{\vec{w}}\left[\left(\tilde{w}(u_j)^2\,\sigma(\langle u_j, x\rangle)^2\right] \leq$

$= -2h(x)^2 + h(x)^2$

$0 \leq \sigma \leq 1$

$|\tilde{w}(u)| \leq A$

$\leq \mathbb{E}_{\vec{w}}[A^2 \cdot 1] = A^2.$

By that fact, and previous inequalities:

$$\mathbb{E}_{u_1,\dots,u_N}\left[\left|\tilde{h}_u(x) - h(x)\right|\right] \leq \frac{1}{N}\sqrt{\sum_{j=1}^{N} A^2} = \frac{1}{N}\sqrt{NA^2} = \frac{A\sqrt{N}}{\sqrt{N}\sqrt{N}} = \frac{A}{\sqrt{N}},$$ so:

$$\mathbb{E}_{u_1,\dots,u_N}\left[\left|\tilde{h}_u(x) - h(x)\right|\right] \leq \frac{A}{\sqrt{N}} \quad \forall x \in \mathbb{R}^d$$ (since this holds for any x),

and:

$$\mathbb{E}_{u_1,\dots,u_N}\left[\tilde{h}(x)\right] = h(x) \quad \text{when } \tilde{w}(u) = \frac{\alpha(u)}{p_u(u)}. \quad \square$$

b) $L(\tilde{h}_u) = \mathbb{E}_{u, x\sim p_x}\left|\tilde{h}_u(x) - h(x)\right| = \mathbb{E}_{x\sim p_x}\left[\mathbb{E}_u\left[\left|\tilde{h}_u(x) - h(x)\right|\right]\right] \leq$

($=$: by definition; $=$: by law of iterated expectations; $\leq$: by a)

$\leq \mathbb{E}_{x\sim p_x}\left[\frac{A}{\sqrt{N}}\right] = \frac{A}{\sqrt{N}}$, so $L(\tilde{h}_u) \leq \frac{A}{\sqrt{N}}$ $\square$.

(2) c) The ANN's architecture is an MLP with 1 hidden layer, and sigmoid activation, and one output layer. In detail:

x: d-dimensional input

U: First linear layer of the MLP, transforms x into N-dimensional space.

σ: Activations applied to outputs of linear layer U, sigmoid applied element-wise.

w: Output layer, single neuron, combines activations.

So the architecture is:

$$x \xrightarrow[\text{input}]{} \underset{(\text{weights } U)}{\text{hidden layer}} \rightarrow \underset{(\text{sigmoid})}{\text{activation}} \rightarrow \underset{(\text{weights } w)}{\text{output layer}} \rightarrow f(x)$$

Total parameters $= \underbrace{N \cdot d}_{U} + \underbrace{N}_{w} = N(d+1)$.

d) $\tilde{h}_w$ is: $\tilde{h}_w(x) = \frac{1}{N}\sum_{j=1}^{N} \tilde{w}(w_j)\sigma(\langle w_j, x\rangle)$, $f(x) = w^T\sigma(Ux) = \sum_{j=1}^{N} w_j \sigma(w_j^T x)$

Since $w_j \sim p_w$ and frozen (as stated), if $w_j = \frac{\tilde{w}(w_j)}{N}$, $\tilde{h}_w(x) = f(x)$.

Then, from a): $|\tilde{w}(w)| \leq A \Rightarrow |w_j| = \left|\frac{\tilde{w}(w_j)}{N}\right| = \frac{|\tilde{w}(w_j)|}{N} \leq \frac{A}{N}$, and so:

$\|w\|_1 = \sum_{j=1}^{N} |w_j| = \sum_{j=1}^{N} \frac{|\tilde{w}(w_j)|}{N} \leq \sum_{j=1}^{N} \frac{A}{N} = N \cdot \frac{A}{N} = A$. So:

$p = 1$, $B = A$ ensure that $\tilde{h}_w$ is a feasible solution to the defined optimization problem.

c) On Piazza, the instructors confirmed asking about empirical loss,

so:

$$\hat{L}(h_w^*) = \frac{1}{n}\sum_{i=1}^{n} |y_i - h_w^*(x_i)| = \frac{1}{n}\sum_{i=1}^{n} |y_i - (w^*)^T \sigma(U x_i)| \leq$$

$$\leq \frac{1}{n}\sum_{i=1}^{n} |y_i - \tilde{h}_w(x_i)| = \underbrace{\hat{L}(\tilde{h}_w)}_{\text{empirical loss}}$$

since w^* is a minimizer, and $\tilde{h}_w$ is a feasible solution.

So:

$$L(h_w^*) = \mathbb{E}_{D,w}\left[\hat{L}(h_w^*)\right] \leq \mathbb{E}_{D,w}\left[\hat{L}(\tilde{h}_w)\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{D,w}\left[|\tilde{h}_w(x) - h(x)|\right]$$

by above

$$\leq \frac{1}{n}\sum_{i=1}^{n} \frac{A}{\sqrt{N}} = \frac{1}{n} \cdot \frac{nA}{\sqrt{N}} = \frac{A}{\sqrt{N}},$$

by result in b), and iid data

so: $L(h_w^*) \leq \frac{A}{\sqrt{N}}$ □.